

River Test Answers

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Ferry travelling across a river(20 points)

1. (4 points) A ferry is crossing a river that is 18m wide at a velocity of 9 ms^{-2} right relative to the water. However, the water is travelling at 3 ms^{-2} down relative to the ground. Calculate:

- (a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is 9.49 ms^{-1} at $S71.57E$

- (b) The time taken to cross the river

Solution: The time taken to cross the river is 2.0s

- (c) The displacement of the boat

Solution: The displacement of the ferry is 18.97m at $S71.57E$

- (d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 6.0m South

2. (4 points) A ferry is crossing a river that is 168m wide at a velocity of 24 ms^{-2} right relative to the water. However, the water is travelling at 8 ms^{-2} up relative to the ground. Calculate:

- (a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is 25.3 ms^{-1} at $N71.57E$

- (b) The time taken to cross the river

Solution: The time taken to cross the river is 7.0s

- (c) The displacement of the boat

Solution: The displacement of the ferry is 177.09m at $N71.57E$

- (d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 56.0m North

3. (4 points) A ferry is crossing a river that is 24m wide at a velocity of 24 ms^{-2} right relative to the water. However, the water is travelling at 4 ms^{-2} down relative to the ground. Calculate:

- (a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is 24.33 ms^{-1} at S80.54E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 1.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 24.33m at S80.54E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 4.0m South

4. (4 points) A ferry is crossing a river that is 8m wide at a velocity of 1 ms^{-2} right relative to the water. However, the water is travelling at 7 ms^{-2} up relative to the ground. Calculate:

(a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is 7.07 ms^{-1} at N8.13E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 8.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 56.57m at N8.13E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 56.0m North

5. (4 points) A ferry is crossing a river that is 24m wide at a velocity of 4 ms^{-2} right relative to the water. However, the water is travelling at 5 ms^{-2} down relative to the ground. Calculate:

(a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is 6.4 ms^{-1} at S38.66E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 6.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 38.42m at S38.66E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 30.0m South

Ferry travelling straight across a river(15 points)

6. (3 points) A boat is crossing a river that is 18m wide with a velocity of 2 ms^{-2} relative to the water. However, the water is travelling at 1 ms^{-2} down relative to the ground. If the boat wants to travel in a straight line across, calculate:

- (a) The resultant velocity of the boat relative to the ground

Solution: The resultant velocity of the ferry is 1.73 ms^{-1} E

- (b) The required angle the boat needs to travel at to cross the river directly across

Solution: The required angle for the ferry to travel at is S63.43E

- (c) The time taken to cross the river

Solution: The time taken for the ferry to cross the river is 10.39 seconds

7. (3 points) A boat is crossing a river that is 35m wide with a velocity of 5 ms^{-2} relative to the water. However, the water is travelling at 1 ms^{-2} up relative to the ground. If the boat wants to travel in a straight line across, calculate:

- (a) The resultant velocity of the boat relative to the ground

Solution: The resultant velocity of the ferry is 4.9 ms^{-1} E

- (b) The required angle the boat needs to travel at to cross the river directly across

Solution: The required angle for the ferry to travel at is N78.69E

- (c) The time taken to cross the river

Solution: The time taken for the ferry to cross the river is 7.14 seconds

8. (3 points) A boat is crossing a river that is 72m wide with a velocity of 9 ms^{-2} relative to the water. However, the water is travelling at 1 ms^{-2} down relative to the ground. If the boat wants to travel in a straight line across, calculate:

- (a) The resultant velocity of the boat relative to the ground

Solution: The resultant velocity of the ferry is 8.94 ms^{-1} E

- (b) The required angle the boat needs to travel at to cross the river directly across

Solution: The required angle for the ferry to travel at is S83.66E

- (c) The time taken to cross the river

Solution: The time taken for the ferry to cross the river is 8.05 seconds

9. (3 points) A boat is crossing a river that is 60m wide with a velocity of 15 ms^{-2} relative to the water. However, the water is travelling at 9 ms^{-2} down relative to the ground. If the boat wants to travel in a straight line across, calculate:

- (a) The resultant velocity of the boat relative to the ground

Solution: The resultant velocity of the ferry is 12.0 ms^{-1} E

- (b) The required angle the boat needs to travel at to cross the river directly across

Solution: The required angle for the ferry to travel at is S59.04E

- (c) The time taken to cross the river

Solution: The time taken for the ferry to cross the river is 5.0 seconds

10. (3 points) A boat is crossing a river that is 9m wide with a velocity of 9 ms^{-2} relative to the water. However, the water is travelling at 4 ms^{-2} up relative to the ground. If the boat wants to travel in a straight line across, calculate:

- (a) The resultant velocity of the boat relative to the ground

Solution: The resultant velocity of the ferry is 8.06 ms^{-1} E

- (b) The required angle the boat needs to travel at to cross the river directly across

Solution: The required angle for the ferry to travel at is N66.04E

- (c) The time taken to cross the river

Solution: The time taken for the ferry to cross the river is 1.12 seconds